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Packaging of a semiconductor device with a plurality of leads

Abstract

A semiconductor device includes a plurality of leads, a semiconductor element electrically connected to the leads and supported by one of the leads, and a sealing resin covering the semiconductor element and a part of each lead. The sealing resin includes a first edge, a second edge perpendicular to the first edge, and a center line parallel to the first edge. The reverse surfaces of the respective leads include parts exposed from the sealing resin, and the exposed parts include an outer reverse-surface mount portion and an inner reverse-surface mount portion that are disposed along the second edge of the sealing resin. The inner reverse-surface mount portion is closer to the center line of the sealing resin than is the outer reverse-surface mount portion. The outer reverse-surface mount portion is greater in area than the inner reverse-surface mount portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 18/301,807, filed Apr. 17, 2023, which is a continuation application of U.S. application Ser. No. 17/472,362, filed Sep. 10, 2021, which is a continuation application of U.S. application Ser. No. 17/109,675, filed Dec. 2, 2020, which is a continuation application of U.S. application Ser. No. 16/242,227, filed Jan. 8, 2019, which is a continuation application of U.S. application Ser. No. 15/416,261, filed Jan. 26, 2017, which claims priority to Japan Patent Application No. 2016-013368 filed Jan. 27, 2016, the entire contents of each of which are incorporated herein by reference, including the original claims.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a semiconductor device.

2. Description of the Related Art

(2) For a semiconductor device including a semiconductor element, typified by a transistor, various configurations have been proposed. An example of a conventional semiconductor device is disclosed in JP-A-2009-71033. The semiconductor device disclosed in this document includes a semiconductor element, a plurality of leads and a sealing resin. The semiconductor element is supported on one of the plurality of leads and electrically connected to the leads. The sealing resin covers the semiconductor element and a part of each lead. The portions of the leads which are exposed from the sealing resin constitute mount portions, which are used for mounting the semiconductor device on e.g. a circuit board. The mount portions are bonded to a circuit board with solder, for example.

(3) In accordance with the specifications of the semiconductor device or the use environment, stress may be generated in the solder for bonding the mount portions and the circuit board. The stress may undesirably cause the solder to crack or peel off.

SUMMARY OF THE INVENTION

(4) The present invention has been proposed under the above circumstances, and an object thereof is to provide a semiconductor device that enhances the mounting strength.

(5) According to an aspect of the invention, there is provided a semiconductor device provided with: a plurality of leads each including an obverse surface and a reverse surface that face away from each other in a thickness direction; a semiconductor element electrically connected to the plurality of leads and supported by the obverse surface of one of the plurality of leads; and a sealing resin covering the semiconductor element and a part of each of the leads. The sealing resin includes a first edge, a second edge and a center line, where the first edge extends along a first direction perpendicular to the thickness direction, the second edge extends along a second direction perpendicular to both the thickness direction and the first direction, and the center line extends in parallel to the first edge. The reverse surfaces of the plurality of leads include a plurality of exposed parts exposed from the sealing resin, and the exposed parts include at least one outer reverse-surface mount portion and at least one inner reverse-surface mount portion that are arranged along the second edge of the sealing resin. The inner reverse-surface mount portion is closer to the center line of the sealing resin than is the outer reverse-surface mount portion, and the outer reverse-surface mount portion is greater in area than the inner reverse-surface mount portion.

(6) With the above arrangements, the outer reverse-surface mount portion, having a relatively large area, can be disposed at the outermost position in the second direction. This feature is advantageous for preventing cracks or other defects from occurring at the mount portion due to large thermal stress.

(7) Further features and advantages of the present invention will become apparent from the following detailed description with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a plan view showing a semiconductor device according to a first embodiment of the present invention;

(2) FIG. 2 is a bottom view showing the semiconductor device according to the first embodiment of the present invention;

(3) FIG. 3 is a plan view showing a main part of the semiconductor device according to the first embodiment of the present invention;

(4) FIG. 4 is an enlarged plan view showing a main part of the semiconductor device according to the first embodiment of the present invention;

- (5) FIG. 5 is an enlarged plan view showing a main part of the semiconductor device according to the first embodiment of the present invention;
- (6) FIG. 6 is a sectional view taken along lines VI-VI in FIG. 3;
- (7) FIG. 7 is a sectional view taken along lines VI-VI in FIG. 3;
- (8) FIG. 8 is a sectional view taken along lines VIII-VIII in FIG. 3;
- (9) FIG. 9 is an enlarged sectional view of a main part, taken along lines VIII-VIII in FIG. 3;
- (10) FIG. 10 is a sectional view taken along lines X-X in FIG. 3;
- (11) FIG. 11 is an enlarged sectional view of a main part, taken along lines X-X in FIG. 3;
- (12) FIG. 12 is an enlarged sectional view of a main part, taken along lines XII-XII in FIG. 3;
- (13) FIG. 13 is a sectional view taken along lines XIII-XIII in FIG. 3;
- (14) FIG. 14 is a schematic plan view showing a step in manufacturing the semiconductor device according to the first embodiment of the present invention;
- (15) FIG. 15 is a schematic plan view showing a step in manufacturing the semiconductor device according to the first embodiment of the present invention;
- (16) FIG. 16 is a graph showing the results of a crack progress test performed on the semiconductor device of the first embodiment and another semiconductor device as a comparative example;
- (17) FIG. 17 is a plan view showing a main part of a variation of the semiconductor device according to the first embodiment of the present invention; and
- (18) FIG. 18 is a plan view showing a main part of another variation of the semiconductor device according to the first embodiment of the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (19) Preferred embodiments of the present invention are described below with reference to the drawings.
- (20) FIGS. 1-13 show a semiconductor device according to a first embodiment of the present invention. The semiconductor device A1 of this embodiment includes a plurality of leads 1, 2 and 3, a semiconductor element 4, and a sealing resin 6.
- (21) FIG. 1 is a plan view showing the semiconductor device A1. FIG. 2 is a bottom view showing the semiconductor device A1. FIG. 3 is a plan view of a main part of the semiconductor device A1. FIG. 4 is an enlarged plan view of a main part of the semiconductor device A1. FIG. 5 is an enlarged plan view of a main part of the semiconductor device A1. FIG. 6 is a sectional view taken along lines VI-VI in FIG. 3. FIG. 7 is an enlarged sectional view of a main part, taken along lines VI-VI in FIG. 3. FIG. 8 is a sectional view taken along lines VIII-VIII in FIG. 3. FIG. 9 is an enlarged sectional view of a main part, taken along lines VIII-VIII in FIG. 3. FIG. 10 is a sectional view taken along lines X-X in FIG. 3. FIG. 11 is an enlarged sectional view of a main part, taken along lines X-X in FIG. 3. FIG. 12 is an enlarged sectional view of a main part, taken along lines XII-XII in FIG. 3. FIG. 13 is a sectional view taken along lines XIII-XIII in FIG. 3. The y direction is a first direction in the present invention, the x direction is a second direction in the present invention, and the z direction is a thickness direction in the present invention.
- (22) The semiconductor device A1 is not limited in size. For example, in this embodiment, the semiconductor device A1 have dimensions of 2.6 to 3.6 mm in the direction x, 2.6 to 3.6 mm in the direction y and 0.7 to 1.0 mm in the direction z.
- (23) The plurality of leads 1, 2 and 3 are electrically connected to the semiconductor element 4, and at least one of them supports the semiconductor element 4. In the illustrated example, the semiconductor element 4 is mounted on the lead 3. In the description below, these leads are referred to as first lead 1, second lead 2 and third lead 3. The first lead 1, the second lead 2 and the third lead 3 may be formed by punching or bending a metal plate, for example. The first lead 1, the second lead 2, and the third lead 3 are made of metal, and preferably, made of Cu, Ni, alloys of Cu or Ni, or 42 alloy, for example. The first lead 1, the second lead 2 and the third lead 3 each may have a thickness of 0.1 to 0.3 mm, and has a thickness of about 0.2 mm in this embodiment.
- (24) As shown in FIG. 3, the first lead 1 and the second lead 2 are arranged side by side in the x

direction. The third lead **3** is spaced apart from the first lead **1** and the second lead **2** in the y direction. As viewed in the z direction, the third lead **3** has the largest dimensions, and the first lead **1** has the smallest dimensions.

(25) As shown in FIGS. **6** and **7**, the first lead **1** has an obverse surface **101** and a reverse surface **102**. The obverse surface **101** and the reverse surface **102** face away from each other in the z direction. As shown in FIGS. **4**, **6** and **7**, the first lead **1** has a first wire bonding portion **111**, a first terminal portion **112** and a first bent portion **115**. The first wire bonding portion **111** is at a position deviated from the first terminal portion **112** in the z direction toward a side which the obverse surface **101** faces. The first wire bonding portion **111** is positioned inward from the first terminal portion **112** in the y direction. In this embodiment, the positional deviation between the first wire bonding portion **111** and the first terminal portion **112** in the z direction is about 0.15 mm. The first bent portion **115** connects the first wire bonding portion **111** and the first terminal portion **112** to each other and has a bent shape as viewed in the x direction.

(26) The first terminal portion **112** has two first end surfaces **121** and one first recessed end surface **122**. The first end surfaces **121** face outward in the y direction. The first recessed end surface **122** is recessed relative to the first end surfaces **121** in the y direction as viewed in the z direction. The first recessed end surface **122** is positioned between the two first end surfaces **121** in the x direction.

(27) As shown in FIG. **2**, the reverse surface of the first terminal portion **112**, which is a part of the reverse surface **102**, constitutes an outer reverse-surface mount portion **150**. As shown in FIG. **7**, the outer reverse-surface mount portion **150** is exposed from the sealing resin **6**. In mounting the semiconductor device **A1** to a circuit board **91**, the outer reverse-surface mount portion **150** is bonded to the circuit board **91** with solder **92**. The outer reverse-surface mount portion **150** has end edges **151** adjoining the first end surfaces **121** and a recessed edge **152** adjoining the first recessed end surface **122**.

(28) As shown in FIGS. **4**, **6** and **7**, the first lead **1** has a first recessed side surface **123** and a first through-hole **130**. The first recessed side surface **123** is recessed in the x direction as viewed in the z direction. The first recessed side surface **123** overlaps with (or has a boundary shared with) the first wire bonding portion **111** and the first bent portion **115**, as viewed in the z direction. The first through-hole **130** penetrates the first lead **1** in the z direction. The first through-hole **130** overlaps with the first bent portion **115**, as viewed in the z direction. The first through-hole **130** also overlaps with the first wire-bonding portion **111** and the first terminal portion **112**, as viewed in the z direction.

(29) The obverse surface **101** is partially covered with a first obverse-surface plating layer **191**. For example, the first obverse-surface plating layer **191** is a Ag-plating layer. In this embodiment, the portion of the obverse surface **101** which constitutes the obverse surfaces of the first wire bonding portion **111** and the first bent portion **115** is covered with the first obverse-surface plating layer **191**.

(30) The reverse surface **102** is covered with a first reverse-surface plating layer **192**. The first recessed end surface **122** is covered with a first side-surface plating layer **193**. The two first end surfaces **121** are exposed without being covered with the first side-surface plating layer **193**. The reverse-surface plating layer **192** and the first side-surface plating layer **193** are integrally formed of a same material. The first obverse-surface plating layer **191** is formed of a material different from that of the first reverse-surface plating layer **192** and first side-surface plating layer **193**. For example, the reverse-surface plating layer **192** and the first side-surface plating layer **193** are a Sn-plating layer.

(31) As shown in FIGS. **8**, **9** and **12**, the second lead **2** has an obverse surface **201** and a reverse surface **202**. The obverse surface **201** and the reverse surface **202** face away from each other in the z direction. As shown in FIGS. **3**, **5**, **8**, **9** and **12**, the second lead **2** has a second wire-bonding portion **211**, a second outer terminal portion **212**, two second inner terminal portions **213** and three second bent portions **215**. The second wire-bonding portion **211** is at a position deviated from the

second outer terminal portion **212** and the second inner terminal portions **213** in the z direction toward a side which the obverse surface **201** faces. The second wire-bonding portion **211** is positioned inward from the second outer terminal portion **212** and the two second inner terminal portions **213** in the y direction. In this embodiment, the positional deviation between the second wire-bonding portion **211** and the second outer and inner terminal portions **212**, **213** in the z direction is about 0.15 mm. Each of the three second bent portions **215** connects the second wire-bonding portion **211** to a corresponding one of the second outer and inner terminal portions **212**, **213** and has a bent shape as viewed in the x direction. The second outer terminal portion **212** is at an outermost position in the x direction. The two second inner terminal portions **213** are positioned inward from the second outer terminal portion **212** in the x direction and aligned with the second outer terminal portion **212** in the x direction. The two second inner terminal portions **213** are positioned between the first terminal portion **112** and the second outer terminal portion **212** in the x direction.

(32) The second outer terminal portion **212** has two second end surfaces **221** and one second recessed end surface **222**. The second end surfaces **221** face outward in the y direction. The second recessed end surface **222** is recessed relative to the second end surfaces **221** in the y direction as viewed in the z direction. The second recessed end surface **222** is positioned between the two second end surfaces **221** in the x direction.

(33) As shown in FIG. 2, the reverse surface of the second outer terminal portion **212**, which is a part of the reverse surface **202**, constitutes an outer reverse-surface mount portion **250**. The outer reverse-surface mount portion **250** is exposed from the sealing resin **9**. In mounting the semiconductor device **A1** to a circuit board **91**, the outer reverse-surface mount portion **250** is bonded to the circuit board **91** with solder **92**, as shown in FIG. 9. The outer reverse-surface mount portion **250** has second end edges **251** adjoining the second end surfaces **221** and a second recessed edge **252** adjoining the second recessed end surface **222**.

(34) As shown in FIG. 2, the reverse surfaces of the second inner terminal portions **213**, which are a part of the reverse surface **202**, constitute two inner reverse-surface mount portions **260**. The inner reverse-surface mount portions **260** are exposed from the sealing resin **6**. In mounting the semiconductor device **A1** to a circuit board **91**, the inner reverse-surface mount portions **260** are bonded to the circuit board **91** with solder **92**.

(35) As shown in FIGS. 5, 8, 9 and 12, the second lead **2** has a second recessed side surface **223** and a second through-hole **230**. The second recessed side surface **223** is recessed in the x direction as viewed in the z direction. The second recessed side surface **223** overlaps with the second wire-bonding portion **211** and the second bent portions **215**, as viewed in the z direction. The second through-hole **230** penetrates the second lead **2** in the z direction. The second through-hole **230** overlaps with the second bent portions **215** as viewed in the z direction. The second through-hole **230** also overlaps with the second wire-bonding portion **211** and the second outer terminal portion **212**, as viewed in the z direction.

(36) As shown in FIG. 3, the obverse surface **201** is partially covered with a second obverse-surface plating layer **291**. For example, the second obverse-surface plating layer **291** is a Ag-plating layer. In this embodiment, the portion of the obverse surface **201** which constitutes the obverse surfaces of the second wire-bonding portion **211** and the second bent portions **215** is covered with the second obverse-surface plating layer **291**.

(37) As shown in FIGS. 9 and 12, the reverse surface **202** is covered with a second reverse-surface plating layer **292**. The second recessed end surface **222** is covered with a second side-surface plating layer **293**. The two second end surfaces **221** are exposed without being covered with the second side-surface plating layer **293**. Also, the end surfaces of the inner reverse-surface mount portions **260** are exposed without being covered with the second side-surface plating layer **293**. The second reverse-surface plating layer **292** and the second side-surface plating layer **293** are integrally formed of a same material. The second obverse-surface plating layer **291** is formed of a

material different from that of the second reverse-surface plating layer **292** and second side-surface plating layer **293**. For example, the second reverse-surface plating layer **292** and the second side-surface plating layer **293** are a Sn-plating layer.

(38) As shown in FIG. 2, the outer reverse-surface mount portion **150** and the outer reverse-surface mount portion **250** are arranged at outermost positions on the opposite sides in the x direction, with the two inner reverse-surface mount portions **260** arranged between the outer reverse-surface mount portions **150** and **250**.

(39) Examples of the dimensions and areas of the outer reverse-surface mount portion **150**, the outer reverse-surface mount portion **250** and the inner reverse-surface mount portions **260** are described below.

(40) Referring to FIG. 2, the dimension L1 in the x direction of the outer reverse-surface mount portions **150**, **250** is about 0.7 mm. The dimension L2 in the x direction of the inner reverse-surface mount portions **260** is about 0.3 mm. The distance S1 between the outer reverse-surface mount portion **150** and the adjacent inner reverse-surface mount portion **260** is equal to the distance S1 between the outer reverse-surface mount portion **250** and the adjacent inner reverse-surface mount portion **260**. In this example, the distance S1 is 0.27 mm. The distance S2 between the two inner reverse-surface mount portions **260** is 0.27 mm, which is equal to the distance S1. In the illustrated example, the dimensional ratio R2 of the dimension L1 to the dimension L2 is 2.33. It is preferable that the dimensional ratio R2 is in a range of 1.7 to 2.5. In this embodiment, all of the outer reverse-surface mount portion **150**, the outer reverse-surface mount portion **250** and the two inner reverse-surface mount portions **260** are equal in dimension in the y direction. The outer reverse-surface mount portion **150** and the outer reverse-surface mount portion **250** are generally in the form of a rectangle elongated in the x direction. Each of the two inner reverse-surface mount portions **260** is in the form of a rectangle that is less elongated than the outer reverse-surface mount portions **150** and **250**.

(41) The outer reverse-surface mount portion **150** and the outer reverse-surface mount portion **250** have the same area E1, which is 0.222 mm² in the illustrated example. The area E2 of each of the inner reverse-surface mount portions **260** is 0.096 mm². In the illustrated example, the area ratio R1 of the area E1 to the area E2 is 2.31. It is preferable that the area ratio R1 is in a range of 1.7 to 2.5.

(42) The ratio R3 of the dimensional ratio R2 to the area ratio R1 is 1.01. It is preferable that the ratio R3 is in a range of 0.68 to 1.47.

(43) As shown in FIGS. 10 and 11, the third lead **3** has an obverse surface **301** and a reverse surface **302**. As shown in FIG. 10, the obverse surface **301** and the reverse surface **302** face away from each other in the z direction. The third lead **3** includes an element bonding portion **311**, a plurality of terminal extensions **312**, and two side extensions **313**. For example, the element bonding portion **311** is rectangular as viewed in the z direction. The semiconductor element **4** is mounted on the element bonding portion **311**. The plurality of terminal extensions **312** extend from the element bonding portion **311** in the y direction and are arranged side by side in the x direction. The two side extensions **313** extend from the element bonding portion **311** toward the opposite sides in the x direction.

(44) As shown in FIG. 2, the portion of the reverse surface **302** which is exposed from the sealing resin **6** constitutes element-side reverse-surface mount portion **350**. In this embodiment, the entirety of the reverse surface **302** constitutes the element-side reverse-surface mount portion **350**. In mounting the semiconductor device A1 to a circuit board **91**, the element-side reverse-surface mount portion **350** is bonded to the circuit board **91** with solder **92**.

(45) As shown in FIGS. 10 and 11, the third lead **3** includes a reverse-side retreated portion **361**, an eave portion **362**, and an obverse-side intermediate end surface **363**.

(46) The reverse-side retreated portion **361** is retreated from the reverse surface **302** at an edge of the third lead **3** as viewed in the z direction. The eave portion **362** is connected to the reverse-side

retreated portion **361** on the obverse surface **301** side in the z direction and projects outward as viewed in the z direction. The obverse-side intermediate end surface **363** connects the obverse surface **301** to the eave portion **362** and is positioned inward from the eave portion **362** as viewed in the z direction. The obverse-side intermediate end surface **363** overlaps with the reverse-side retreated portion **361** as viewed in the thickness direction.

(47) In this embodiment, the reverse-side retreated portion **361**, the eave portion **362**, and the obverse-side intermediate end surface **363** are provided at an edge of the third lead **3** which is closer to the first and the second leads **1** and **2** as viewed in the z direction, and at opposite edges of the third lead **3** in the x direction, and at an edge of the third lead **3** which is opposite to the leads **1**, **2** between the terminal extensions **312**.

(48) As shown in FIG. **3**, the third lead **3** has a plurality of obverse-side recesses **371**. The obverse-side recesses **371** are provided at positions avoiding the semiconductor element **4** as viewed in the z direction. The obverse-side recesses **371** are recessed from the obverse surface **301** in the thickness direction. In this embodiment, the obverse-side recesses **371** are provided at respective root portions of the terminal extensions **312** and two side extensions **313**.

(49) As shown in FIG. **11**, the obverse surface **301** of the third lead **3** is covered with a third obverse-surface plating layer **391**. Specifically, in the illustrated example, the third obverse-surface plating layer **391** covers the obverse surface **301** except the obverse surfaces of the terminal extensions **312**. For example, the third obverse-surface plating layer **391** is a Ag-plating layer.

(50) The reverse surface **302** is covered with a third reverse-surface plating layer **392**. The side surface of the third lead **3** is covered with a third side-surface plating layer **393** except the end surfaces of the terminal extensions **312** and the end surfaces of the two side extensions **313**. The third reverse-surface plating layer **392** and the third side-surface plating layer **393** are integrally formed of a same material. The third obverse-surface plating layer **391** is formed of a material different from that of the third reverse-surface plating layer **392** and third side-surface plating layer **393**. For example, the third reverse-surface plating layer **392** and the third side-surface plating layer **393** are a Sn-plating layer.

(51) The semiconductor element **4** is an element that performs electrical functions of the semiconductor device **A1**. The type of semiconductor element **4** is not particularly limited. As shown in FIGS. **3** and **6**, in this embodiment, the semiconductor element **4** is configured as a transistor. The semiconductor element **4** includes an element body **40**, a first electrode **41**, a second electrode **42** and a third electrode **43**.

(52) The first electrode **41** and the second electrode **42** are arranged on the obverse surface of the element body **40**. The third electrode **43** is arranged on the reverse surface of the element body **40**. In this embodiment, the first electrode **41** is a gate electrode, the second electrode **42** is a source electrode, and the third electrode **43** is a drain electrode.

(53) The semiconductor device **A1** has a first wire **51** and a plurality of second wires **52**. The first wire **51** is connected to the first electrode **41** and the first wire-bonding portion **111** of the first lead **1**. The second wires **52** are connected to the second electrode **42** and the second wire-bonding portion **211** of the second lead **2**.

(54) The third electrode **43** is mounted to the element bonding portion **311** of the third lead **3** via a conductive bonding material **49**. Specifically, the third electrode **43** is bonded to the third obverse-surface plating layer **391** on the obverse surface **301** of the element bonding portion **311** with the conductive bonding material **49**.

(55) The sealing resin **6** covers the semiconductor element **4**, the first wire **51**, the second wires **52** and a part of each of the first lead **1**, the second lead **2** and the third lead **3**. For example, the sealing resin **6** is made of a black epoxy resin.

(56) As shown in FIGS. **1**, **2** and **6**, the sealing resin **6** has a sealing-resin obverse surface **61**, a sealing-resin reverse surface **62** and a sealing-resin side surface **63**. The sealing-resin obverse surface **61** and the sealing-resin reverse surface **62** face away from each other in the z direction.

The sealing-resin obverse surface **61** faces the same side as the obverse surfaces **101**, **201** and **301**. The sealing-resin reverse surface **62** faces the same side as the reverse surfaces **102**, **202** and **302**. The sealing-resin side surface **63** is connected to the sealing-resin obverse surface **61** and the sealing-resin reverse surface **62** and slightly inclined with respect to the z direction.

(57) All of the outer reverse-surface mount portions **150** and **250**, the two inner reverse-surface mount portions **260** and the element-side reverse-surface mount portion **350** are exposed from the sealing resin **6**. The outer reverse-surface mount portions **150** and **250**, the two inner reverse-surface mount portions **260** and the element-side reverse-surface mount portion **350** are flush with the sealing-resin reverse surface **62** of the sealing resin **6**.

(58) FIG. **14** shows a lead frame **10** used for making the semiconductor device **A1**. The lead frame **10** is a metal plate including portions to become the first lead **1**, the second lead **2** and the third lead **3**.

(59) The portions of the lead frame **10** which are to become the obverse surfaces **101**, **201** and **301** are provided with a Ag-plating layer, which is to become the first obverse-surface plating layer **191**, the second obverse-surface plating layer **291** and the third obverse-surface plating layer **391**. The portions of the lead frame **10** which are to become the reverse surfaces **102**, **202** and **302** are provided with a Sn-plating layer, which is to become the first reverse-surface plating layer **192**, the second reverse-surface plating layer **292** and the third reverse-surface plating layer **392**. The side surface of the lead frame **10** along the z direction is provided with a Sn plating layer, which is to become the first side-surface plating layer **193**, the second side-surface plating layer **293** and the third side-surface plating layer **393**.

(60) As shown in FIG. **15**, the semiconductor element **4** is mounted on the lead frame **10**. Then, the first wire **51** and the second wires **52** are bonded. Then, the sealing resin **6** is formed. Then, the lead frame **10** is cut along the cutting lines **81**, **82** and **83**. This cutting operation provides the first end surfaces **121** of the first terminal portion **112**, the second end surfaces **221** of the second outer terminal portion **212**, the end surfaces of the terminal extensions **312** and the end surfaces of the side extensions **313**. These surfaces are not provided with the first side-surface plating layer **193**, the second side-surface plating layer **293** or the third side-surface plating layer **393**.

(61) The advantages of the semiconductor device **A1** are described below.

(62) According to this embodiment, the outer reverse-surface mount portions **150** and **250** positioned at the outermost positions in the x direction have a larger area than the inner reverse-surface mount portions **260** located at inner positions in the x direction. According to a study by the inventors, a durability test in which the semiconductor device **A1** mounted to a circuit board **91** or the like with solder **92** is alternately subjected to a high temperature state and a low temperature state showed that a large stress is generated in the solder **92** located at the outermost positions in the x direction. With the semiconductor device **A1**, the outer reverse-surface mount portions **150** and **250**, which are located at the outermost positions, have a relatively large area, whereby cracking or the like due to such a large stress is prevented. Thus, the semiconductor device **A1** has enhanced mounting strength.

(63) In the semiconductor device **A1**, the outer reverse-surface mount portion **150** and the outer reverse-surface mount portion **250** are provided at opposite ends in the x direction. This prevents cracking of the solder **92** in a balanced manner at opposite ends in the x direction.

(64) FIG. **16** shows the results of a crack progress test performed on the semiconductor device **A1** and another semiconductor device as a comparative example. In this test, the semiconductor devices mounted to a circuit board were repetitively subjected to a temperature change cycle between a low temperature (−55 degrees Celsius) and a high temperature (150 degrees Celsius), and the progress of cracking at the solder was examined. The horizontal axis represents the number of the temperature change cycles, whereas the vertical axis represents the degree of crack progress, where the crack causing complete release of the solder is defined as 100%. As the comparative example, use was made of a semiconductor device having a plurality of reverse-surface mount

portions which were similar in structure to the inner reverse-surface mount portions **260** and equal in size to each other. In other words, the area ratio **R1** of the comparative example was 1.0. As shown in the figure, the crack progress of the comparative example was 53% after 1000 cycles and reached 100% after 1765 cycles. On the other hand, for the outer reverse-surface mount portions **150** and **250** of the semiconductor device **A1**, the crack progress was only 18% after 1000 cycles and 59% after 2000 cycles. The crack progress reached 100% after 3900 cycles. In this way, the semiconductor device **A1** significantly suppresses the progress of solder cracking, as compared to the comparative example. It is desirable that the crack progress reaches 100% after at least 3000 cycles, which is found to be achievable when the area ratio **R1** is 1.7. When the area ratio **R1** or the dimensional ratio **R2** exceeds 2.5, the outer reverse-surface mount portions **150** and **250** occupy an excessively large area of the semiconductor device **A1**, which is not desirable for proper layout of the reverse-surface mount portions.

(65) Thus, it is desirable that the area ratio **R1** of the outer reverse-surface mount portion **150** or **250** to the inner reverse-surface mount portions **260** is in a range of 1.7 to 2.5. With such a ratio, cracking of solder **92** is reliably prevented while an increase in size of the semiconductor device **A1** due to an excessively large size of the outer reverse-surface mount portions **150** and **250** is avoided.

(66) The area ratio **R1** in the above-described range is realized when the dimensional ratio **R2** of the dimension in the x direction of the outer reverse-surface mount portion **150** or **250** to the dimension in the x direction of the inner reverse-surface mount portions **260** is in a range of 1.7 to 2.5. With the dimensional ratio **R2** in such a range, the dimensions in the y direction of the outer reverse-surface mount portions **150** and **250** are prevented from becoming too large. In this embodiment, the outer reverse-surface mount portions **150** and **250** are equal in dimension in the y direction to the inner reverse-surface mount portions **260** and do not project in the y direction relative to the inner reverse-surface mount portions **260**.

(67) To achieve the area ratio **R1** and the dimensional ratio **R2** in the above-described range, it is desirable that the ratio **R3** of the dimensional ratio **R2** to the area ratio **R1** is in a range of 0.68 to 1.47.

(68) As described above, the first terminal portion **112** has two first end surfaces **121** and one first recessed end surface **122**, as shown in FIGS. 4 and 7. The first end surfaces **121** are provided by the cutting process described with reference to FIG. 15, and the lead **1** is exposed at the first end surfaces **12**. On the other hand, the first recessed end surface **122** is covered with the first side-surface plating layer **193**. The first side-surface plating layer **193**, which may be a Sn-plating layer, has a higher wettability to the solder **92** than the first lead **1** has. Thus, in bonding the outer reverse-surface mount portion **150** to the circuit board **91** with solder **92**, the solder **92** covers the first side-surface plating layer **193** (first recessed end surface **122**) as well. This further enhances the mounting strength. The first side-surface plating layer **193** is integral with and formed of a same material as the first reverse-surface plating layer **192**. This allows the solder **92** to integrally adhere to the outer reverse-surface mount portion **150** (first reverse-surface plating layer **192**) and the first recessed end surface **122** (first side-surface plating layer **193**). This is favorable for enhancing the mounting strength.

(69) As described above, the second outer terminal portion **212** has two second end surfaces **221** and one second recessed end surface **222**, as shown in FIGS. 5 and 9. The second end surfaces **221** are provided by the cutting process described with reference to FIG. 15, and the second lead **2** is exposed at the second end surfaces **221**. On the other hand, the second recessed end surface **222** is covered with the second side-surface plating layer **293**. The second side-surface plating layer **293**, which may be a Sn-plating layer, has a higher wettability to the solder **92** than the second lead **2** has. Thus, in bonding the outer reverse-surface mount portion **250** to the circuit board **91** with solder **92**, the solder **92** covers the second side-surface plating layer **293** (second recessed end surface **222**) as well. This further enhances the mounting strength. The second side-surface plating layer **293** is integral with and formed of a same material as the second reverse-surface plating layer

292. This allows the solder **92** to integrally adhere to the outer reverse-surface mount portion **250** (second reverse-surface plating layer **292**) and the second recessed end surface **222** (second side-surface plating layer **293**). This is favorable for enhancing the mounting strength.

(70) A larger current flows through the second electrode **42**, which is a source electrode, than through the first electrode **41**, which is a gate electrode. The second lead **2**, through which a large current flows, is provided with the second outer terminal portion **212** and the two second inner terminal portions **213**, which is favorable for reducing the resistance.

(71) The provision of the first recessed side surface **123** and the first through-hole **130** enhance the bonding strength between the first lead **1** and the sealing resin **6**. The provision of the second recessed side surface **223** and the second through-hole **230** enhances the bonding strength between the second lead **2** and the sealing resin **6**. The provision of the reverse-side retreated portion **361**, the eave portion **362**, the obverse-side intermediate end surface **363** and the obverse-side recesses **371** enhances the bonding strength between the third lead **3** and the sealing resin **6**.

(72) FIGS. **17** and **18** show variations of the semiconductor device **A1**. In these figures, the elements that are identical or similar to those of the above example are designated by the same reference signs as those used for the above example.

(73) In the variation shown in FIG. **17**, the first terminal portion **112** has one first end surface **121** and two first recessed end surfaces **122**. The first end surfaces **121** is positioned between the two first recessed end surface **122** in the x direction. The first terminal portion **112** having such a configuration is formed by using a lead frame **10** indicated by imaginary lines in the figure. In this variation again, the two first recessed end surfaces **122** are covered with the first side-surface plating layer **193**. Thus, this variation also enhances the bonding strength of the semiconductor device **A1**.

(74) In the variation shown in FIG. **18**, the first terminal portion **112** has one first end surface **121** and one first recessed end surface **122**. The first end surface **121** and the first recessed end surface **122** are arranged side by side in the x direction. The first terminal portion **112** having such a configuration is formed by using a lead frame **10** indicated by imaginary lines in the figure. In this variation again, the first recessed end surface **122** is covered with the first side-surface plating layer **193**. Thus, this variation also enhances the bonding strength of the semiconductor device **A1**.

(75) It is to be noted that the configurations of the variations shown in FIGS. **17** and **18** are also applicable to the second end surfaces **221** and the second recessed end surface **222** of the second wire-bonding portion **211** of the second lead **2**.

(76) The semiconductor device according to the present invention is not limited to the embodiments described above. Various design changes can be made to the specific configurations of the elements of the semiconductor device according to the present invention.

Claims

1. A semiconductor device comprising: a first lead; a semiconductor element spaced apart from the first lead in a first direction in plan view and electrically connected to the first lead; a second lead on which the semiconductor element is provided; and a sealing resin covering the semiconductor element and a part of the first lead, wherein the sealing resin comprises a first resin side surface extending in a second direction perpendicular to the first direction in plan view, the first lead comprises a protrusion, a bottom-surface exposed portion, and an inner portion, the protrusion projecting out from the first resin side surface in the first direction, the inner portion being embedded within the sealing resin, the bottom-surface exposed portion connects the protrusion and the inner portion to each other, the inner portion of the first lead comprises a wire bonding part and a bent part, the bent part connecting the bottom-surface exposed portion and the wire bonding part to each other, and in a cross section, the second lead is formed with an eave portion extending toward the first lead in the first direction, the eave portion including a curved surface at an upper

- side of the eave portion, the eave portion being formed in an area between an upper surface of the second lead and a lower surface of the second lead, and the first lead is formed with openings each extending from an edge of the first lead to the inner portion of the first lead in plan view.
2. The semiconductor device according to claim 1, wherein the eave portion includes a flat surface at a lower side of the eave portion.
 3. The semiconductor device according to claim 2, wherein the flat surface of the eave portion has a greater length in the first direction than the curved surface of the eave portion.
 4. The semiconductor device according to claim 3, wherein the upper surface of the second lead is formed with a recess, and the semiconductor element is disposed between the recess and the eave portion in plan view.
 5. The semiconductor device according to claim 4, wherein part of the sealing resin is disposed within the recess, and the sealing resin covers the flat surface of the eave portion.
 6. The semiconductor device according to claim 5, further comprising a third lead disposed next to the first lead, wherein the third lead is electrically connected to a gate electrode of the semiconductor element.
 7. The semiconductor device according to claim 6, wherein the first resin side surface is inclined with respect to the first direction.
 8. The semiconductor device according to claim 7, wherein the second lead includes a partially widened portion that is greater in length in the second direction than an adjacent portion of the second lead.
 9. The semiconductor device according to claim 8, wherein at least a part of the wire bonding part of the inner portion of the first lead is disposed higher than the upper surface of the second lead.
 10. The semiconductor device according to claim 9, wherein the first lead comprises a recessed end surface and two end face portions flanking the recessed end surface, each of the end face portions being not flush with the recessed end surface.
 11. The semiconductor device according to claim 10, wherein each of the two end face portions is smaller in size along the second direction than the recessed end surface.
 12. The semiconductor device according to claim 2, wherein the wire bonding part of the first lead comprises an upper surface and a lower surface that is higher than the flat surface of the eave portion.
 13. The semiconductor device according to claim 1, wherein the sealing resin comprises a lower surface, an upper surface and a plurality of side surfaces including said first resin side surface, the plurality of side surfaces as a whole tapering from the lower surface of the sealing resin toward the upper surface of the sealing resin.
 14. The semiconductor device according to claim 13, wherein the lower surface of the second lead is exposed from the lower surface of the sealing resin.
 15. The semiconductor device according to claim 1, wherein the upper surface of the second lead comprises a region exposed from the sealing resin.
 16. The semiconductor device according to claim 1, wherein the bent part of the first lead comprises an upper inclined surface that is smaller in inclination angle relative to the first direction than the first resin side surface of the sealing resin.
 17. The semiconductor device according to claim 1, wherein the second lead comprises a flat end portion that is disposed above the eave portion and arranged to face the wire bonding part of the first lead via part of the sealing resin.
 18. The semiconductor device according to claim 1, wherein the sealing resin comprises a second resin side surface opposite to the first resin side surface in the first direction, the eave portion being disposed closer to the first resin side surface than to the second resin side surface in the first direction.
 19. The semiconductor device according to claim 1, wherein the eave portion is elongated in the second direction and greater in length in the second direction than the first lead.

20. The semiconductor device according to claim 1, wherein the eave portion is elongated in the second direction and greater in length in the second direction than the semiconductor element, the semiconductor device further comprising a wire electrically connecting the semiconductor element and the wire bonding part of the first lead to each other, an entirety of the wire is embedded in the sealing resin.
